

Two challenges, one laser

Entangling atoms, then making them compute —
validated on Pasqal Cloud and run twice on the real quantum computer

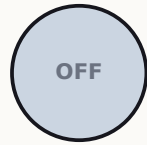
Challenge 01

F: 0.75 → 1.000000

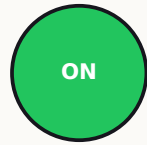
Challenge 02

P_MIS: 0.66 → 0.999999

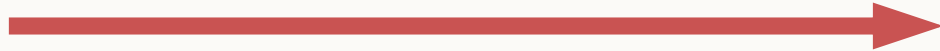
The machine, in thirty seconds



$|g\rangle = \text{OFF}$



$|r\rangle = \text{ON}$



one laser shines on ALL atoms at once

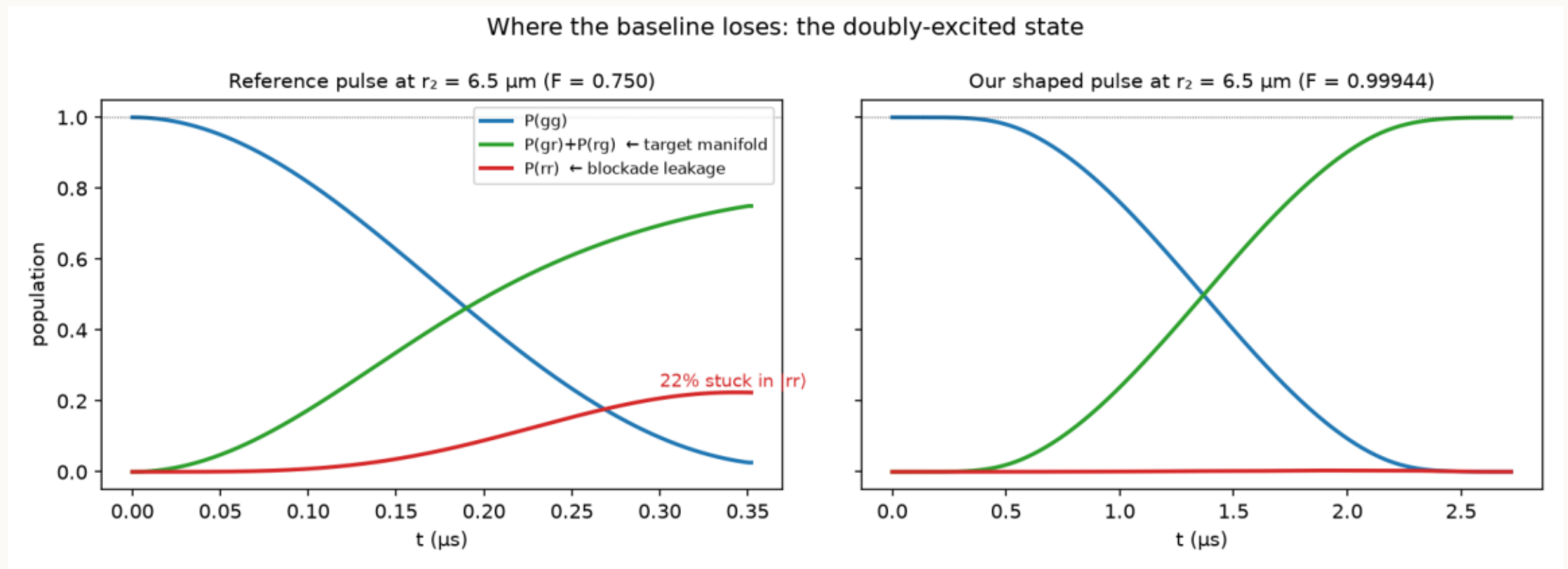
- Every atom is a light switch: ground state $|g\rangle$ is OFF, Rydberg state $|r\rangle$ is ON.
- One laser drives all atoms together — we control its power $\Omega(t)$ and its frequency offset $\delta(t)$ over time.
- Two excited atoms repel with energy $V = C_6/r^6$ — so close atoms can never both be ON. That is the blockade.
- Everything in this talk is built from those three facts.

Challenge 1 — entangle two atoms

- Take two atoms, both OFF, and one laser pulse. Produce the Bell state $|\Psi^+\rangle = (|gr\rangle + |rg\rangle)/\sqrt{2}$ — exactly one atom ON, but genuinely shared between both. That is entanglement.
- We are scored on fidelity, $F = |\langle\Psi^+|\psi(T)\rangle|^2$ — how close our final state is to the ideal; 1.0 is perfect.
- The twist: do it at two different spacings, 5.0 and 6.5 micrometers, beating the starter pulse at both.
- The starter pulse scores 0.9926 at the close spacing — but only 0.7500 at the far one. That gap is the whole challenge.

Why it breaks: blockade quality is V/Ω — interaction over drive. At 6.5 μm it drops to 1.8, and 22% of the population leaks into the forbidden both-ON state $|rr\rangle$.

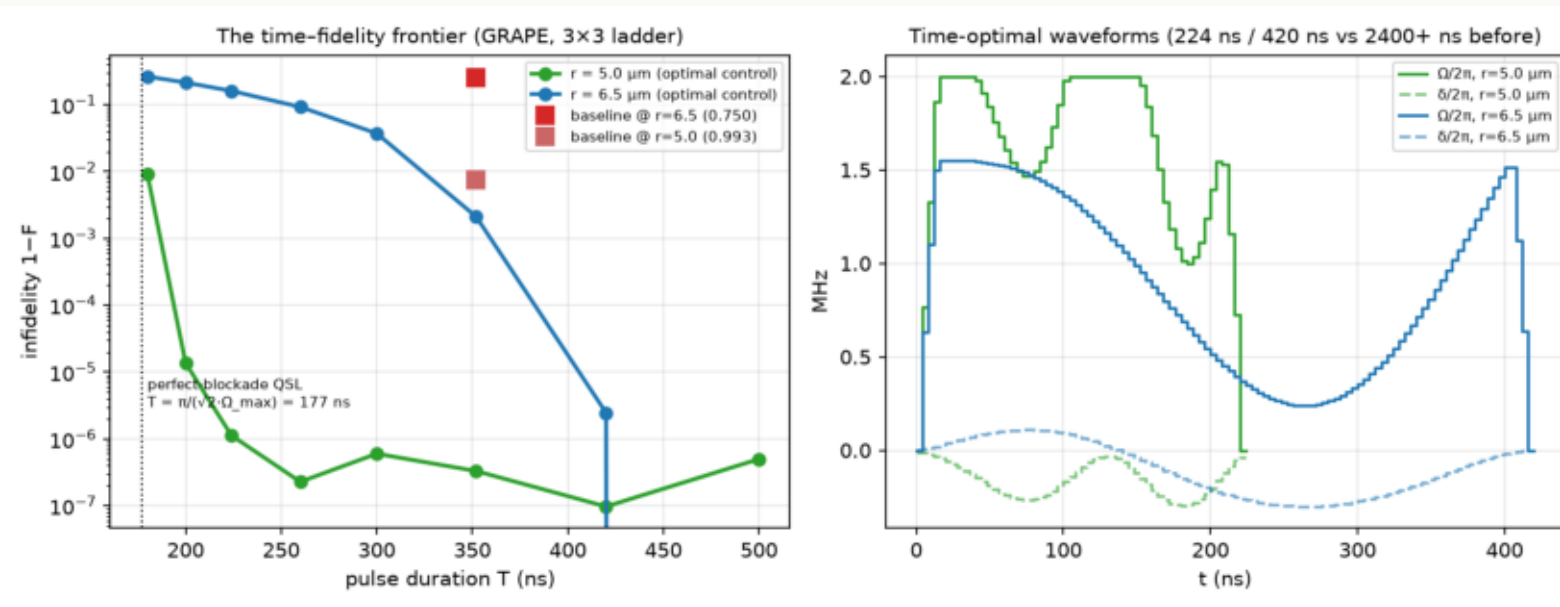
Watch the baseline fail — then watch ours not



x-axis: time during the pulse (μs) · y-axis: probability of each two-atom configuration

- On the left, the baseline: the red curve — the forbidden $|rr\rangle$ state — fills to 22% and the green target stalls at 0.75.
- On the right, our pulse: we lower Ω to restore $V/\Omega \geq 9$, add smooth ramps and a small δ that cancels the $\Omega^2/2V$ energy shift — red stays at zero, green reaches 1.0.

Then we found the speed limit



left — x: pulse duration (ns), y: error $1-F$ on a log scale · right — the winning waveforms

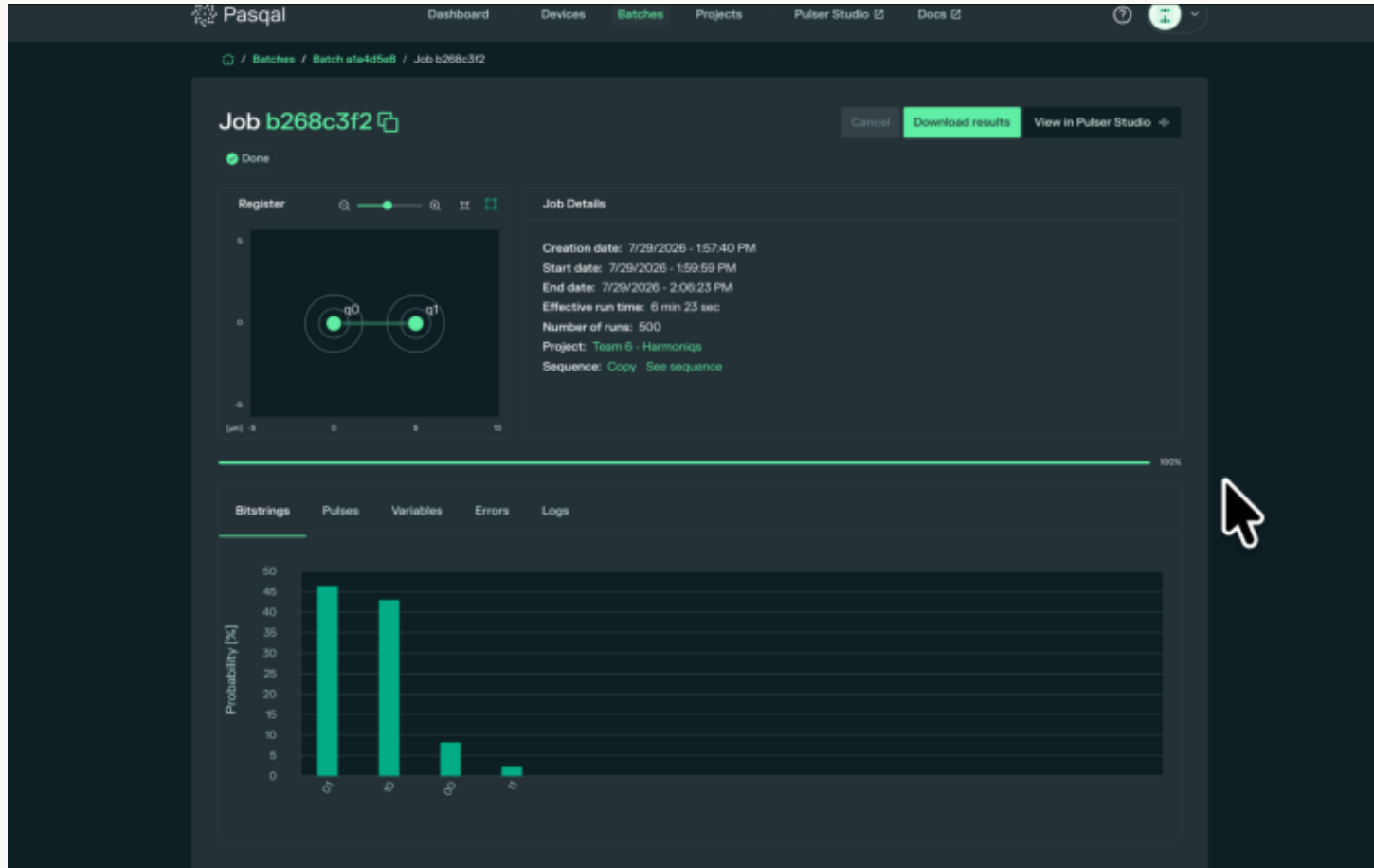
Theory sets a floor: no pulse can beat $T = \pi/(\sqrt{2} \cdot \Omega_{\text{max}}) = 177 \text{ ns}$.

We reach $F = 0.999999$ at 224 ns — just 25 ns above that floor.

The full 16-point frontier took 5.4 seconds of laptop compute: an exact 3-state model with analytic gradients.

Why speed matters: under real noise, slow pulses collapse ($0.999 \rightarrow 0.73$) while short ones keep 0.93+. Duration IS the noise coupling.

500 shots on real atoms: 89.4% entangled



We sent our 260-nanosecond pulse to the real machine.

The two tall bars ARE the Bell state: '01' at 46% plus '10' at 43% — 89.4% with exactly one atom ON.

The forbidden '11' shows just 2.4%: the blockade held on real atoms.

We published the expected window BEFORE the run — and hit it.

Bonus: the portal labels our pulse area $5\pi/7$ — that is $\pi/\sqrt{2}$, the two-atom collective enhancement, on the machine's own dashboard.

Pasqal's own portal · x: measured two-atom outcome · y: % of 500 shots · FRESNEL_CAN1, France

Same laser, bigger question: can the atoms compute?

Challenge 1 taught us three things we carry forward:

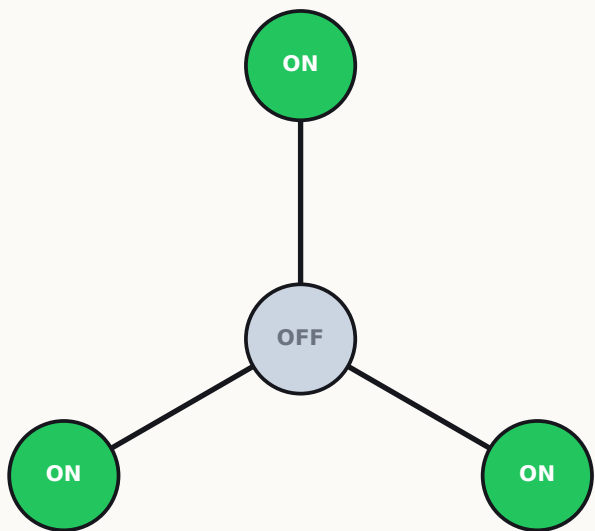
- The blockade is not a nuisance — it is a CONSTRAINT the hardware enforces for free.
- Short, smooth pulses win on real hardware, because noise accumulates with time.
- Verify everything: one scorer, predictions published before runs, every number traceable.

Challenge 2 uses the blockade as a computer: place atoms so the physics itself encodes a hard puzzle — then one laser sweep makes a photograph reveal the answer.

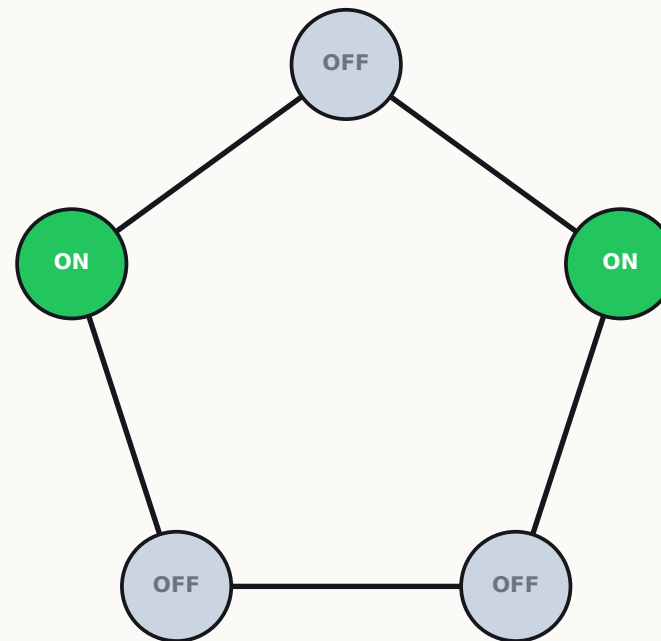
Challenge 2 — the seating puzzle, played by nature

Maximum Independent Set: seat the most people with no two neighbors — a textbook NP-hard problem.

Atoms closer than 7.2 μm cannot both be ON, so atom positions ARE the graph, and physics enforces the rule.



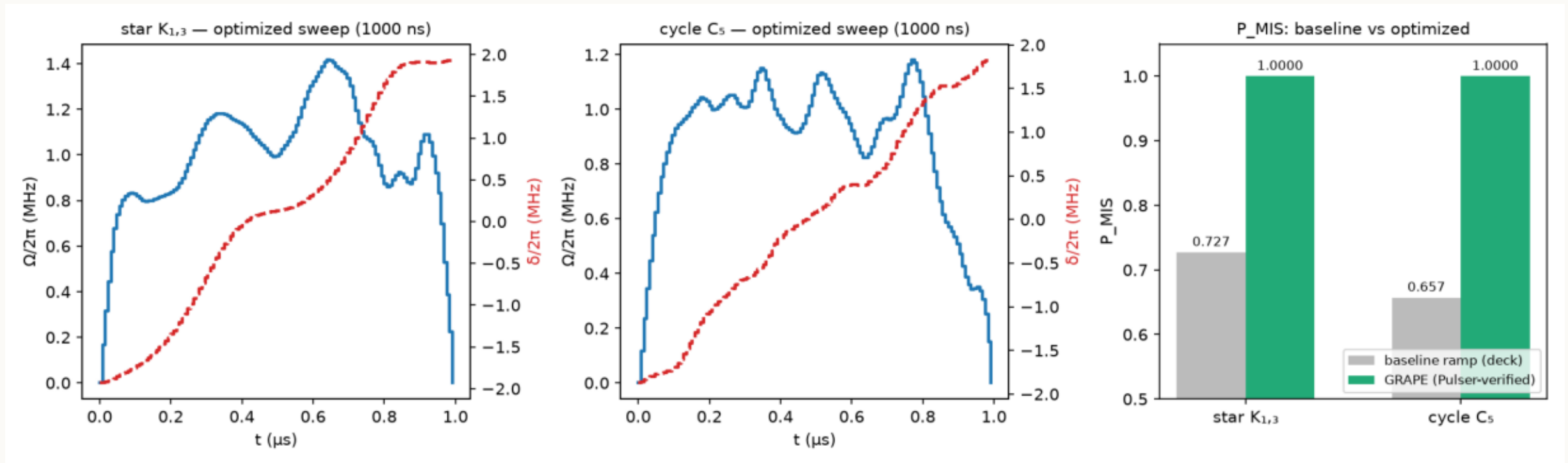
STAR graph: the best answer is all 3 leaves ON, hub OFF — unique.



PENTAGON: any 2 non-neighbors ON — five equally good answers.

Score: P_MIS — the chance one photograph shows a best answer. Baselines to beat: 0.727 (star), 0.657 (pentagon).

Our sweep: near-perfect on both puzzles



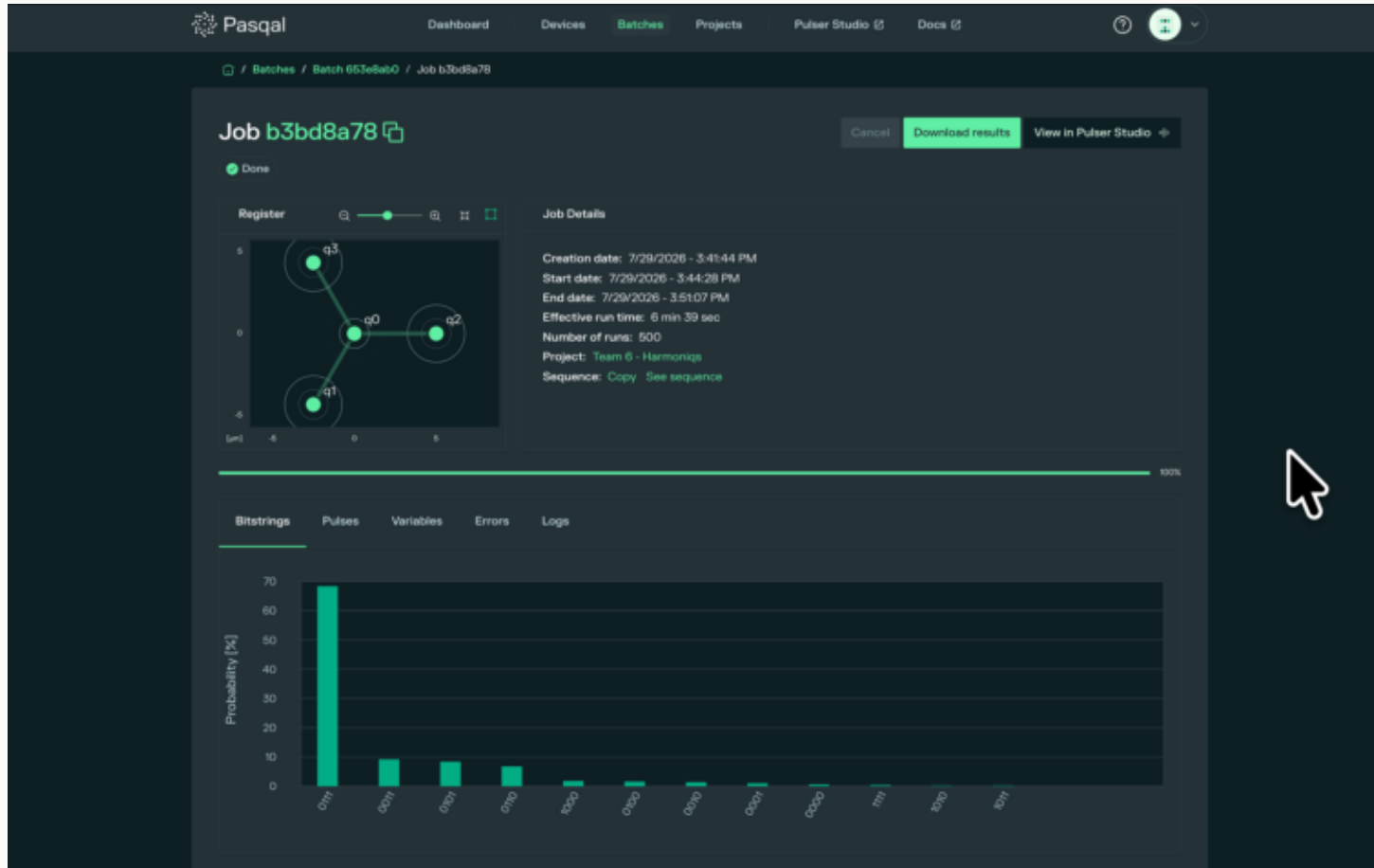
- left/middle — x: time (μs), blue: laser power Ω , red: frequency offset δ · right — x: puzzle, y: P_{MIS}
Instead of the baseline's slow careful ramp, we let a gradient optimizer shape both laser knobs to maximize the answer probability directly — and for the pentagon we target ALL FIVE valid answers at once.

— Team <https://github.com/roshaninfordham/quantum-hackathon>
Result: 0.727 → 0.999998 on the star, 0.657 → 0.999999 on the pentagon — and our sweeps are 4× shorter.

1500 photographs, 1500 correct — then we simplified

- On Pasqal's cloud, every one of 1500 shots across three sweeps showed a correct answer.
- The pentagon's five valid answers came back 114, 98, 97, 96 and 95 times — statistically even. The machine holds all five answers at once and samples them fairly. That is visibly quantum.
- Then the simplification: just FIVE knobs — one power bump, a tilt, and two sine corrections — reach 0.996–0.999, found in thirty seconds, with bandwidth under 1 MHz by construction.
- Those five knobs, reused UNCHANGED on bigger rings, still beat the baseline (+0.30 on the 9-ring). On a structurally different graph they lose — an honest limit, and the map for Challenge 3.

Real atoms pick the right answer, 7 to 1



Look at the register panel: the hub linked to three leaves — Pasqal's own UI is showing the graph we encoded in atom positions.

The towering bar is 0111 — the correct answer — at 68.4%. Second place gets 9.2%.

It landed below our predicted 80%: holding three fragile ON-atoms compounds readout loss. Prediction published first, analysis after.

Total quantum time for all 500 shots: half a millisecond.

Pasqal's portal · top-left: OUR STAR, drawn by the machine's register viewer · bottom — x: pattern, y: % of 500 shots

The score sheet

Challenge 1 — Bell-state fidelity F

	$r_1 = 5.0 \mu\text{m}$	$r_2 = 6.5 \mu\text{m}$	duration
Baseline	0.9926	0.7500	352 ns
Ours (best)	1.000000	1.000000	224-600 ns
Real QPU	89.4% entangled (500 shots)	n/a on lattice	260 ns

Challenge 2 — answer probability P_{MIS}

	star $K_{1,3}$	pentagon C_5	duration
Baseline	0.727	0.657	4000 ns
Ours (best)	0.999998	0.999999	1000-2000 ns
Cloud	500/500	500/500 + 500/500	—
Real QPU	68.4% exact (7× runner-up)	n/a on lattice	1000 ns

Both stages completed. Every number regenerates from the public repo; every hardware claim has a batch ID.

What we're taking home

- A physics diagnosis beats a black-box search: one number, V/Ω , explained both baseline failures.
- On real hardware, pulse duration is the noise coupling — we measured the ranking flip, then designed for it.
- Five interpretable knobs get you within half a percent of full optimal control — and they transfer across a graph family at zero cost. That is the recipe that scales to Challenge 3's 80-atom instances.
- Discipline compounds: one scorer per challenge, predictions published before every hardware run, failures analyzed in the open. Two QPU runs, both informative — one confirmed, one taught us SPAM.
- Thanks to the Pasqal stack: Pulser's device models, the free cloud emulator, and FRESNEL_CAN1 — whose portal drew our graph back at us.